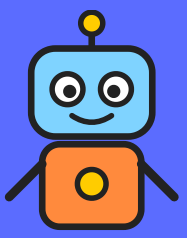
Robotics Project #9 · Kids.JoinDev

micro:bit Night-Light Robot

Ages 8+ · 40 min · Build with a grown-up!



🎯 Goal

Make a robot whose eyes light up by themselves when it gets dark.

🧰 You need



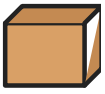
BBC micro:bit



USB cable



AAA battery pack



cardboard robot head



tracing paper

🔧 Build it

1. Open python.microbit.org with a grown-up. ☐
2. Copy the night-light code and send it to the micro:bit. ☐
3. Cut a window in the robot head and tape tracing paper over it. ☐
4. Put the micro:bit behind the window. ☐
5. Cover it with your hand: the lights come on! ☐
6. Change the number 60 to make it more or less sensitive. ☐

⚠️ Safety

Use the battery pack. Grown-up helps cut the window.

💡 How it works

The LED screen can also measure light. The code checks it again and again: dark means lights on (if... else...).
Code: night_light.py (in the project folder)

github.com/dev-repos/kids-robotics-projects · youtube.com/@Kids.JoinDev · CC BY 4.0